

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

YEARLY EXAMINATION

2006

MATHEMATICS
EXTENSION 1

*Time Allowed - Two hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- * Approved calculators may be used.
- * There are 7 questions of equal value. Start each question on a new page.
- * Show all necessary working. Marks may be deducted otherwise.

Begin each question on a new page

Question 1

- a) Solve $16^{4-x} = \frac{1}{8^x}$. 3
- b) Solve $\frac{x-6}{x} \geq 3$ 3
- c) Find the exact value of the gradient of the tangent to $y = \frac{1}{\sqrt{2x-4}}$ at the point where $x = 3$. 3
- d) Find the coordinates of the point P which divides the interval joining points A(4,6) and B(13,5) externally in the ratio 4:1. 2
- e) Evaluate $\sum_{k=3}^6 (-1)^k k$ 1

Question 2

- a) Find the acute angle between the lines $y = 2x - 1$ and $3x - 2y + 1 = 0$. Answer correct to the nearest degree 3
- b) Solve $\sin 2\theta = -\sqrt{3} \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$ 3
- c) Solve $x^4 + 2x^2 - 8 = 0$ 3
- d) Find, from first principles, the derivative of $y = x^2 - 2x$ 3

Question 3

- a)

The line $3x + 4y - k = 0$ is a tangent to the circle $x^2 + y^2 = 16$. Find the possible values of k .

3
- b)

(i)

On the same axes, sketch the graphs of $y = |x|$ and $y = (x - 2)^2$

2

(ii)

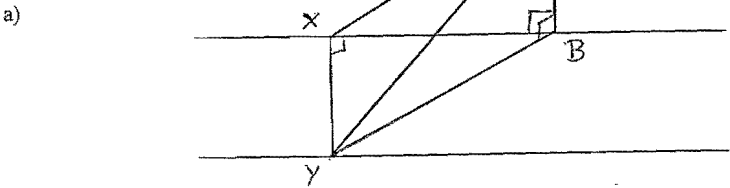
Hence solve $|x| < (x - 2)^2$

3
- c)

The curve $y = x^3 + ax^2 + bx + c$ has a point of inflexion at $(1, 2)$ and a stationary point when $x = 4$. Find the value of a , b and c .

4

Question 4



From a point X on a canal bank, a surveyor measures the angle of elevation of a tower 50m high and on the same bank to be 56° . From a point Y on the other bank and exactly opposite X , another surveyor finds the angle of elevation of the tower is 39° . Assuming that X , Y and the foot of the tower are all on the same level, show that the breadth of the canal is given by

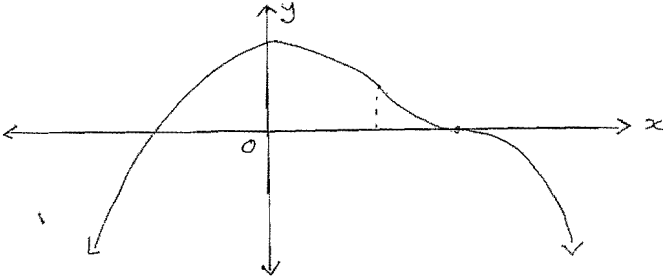
$$b = 50\sqrt{\cot^2 39^\circ - \cot^2 56^\circ}$$

and evaluate this expression, giving your answer correct to 3 significant figures.

3

- b)
- Write the expansion for $\cos(A - B)$. Hence find the exact value for $\cos 15^\circ$.
- 3

- c)
- The diagram shows a function $y = f(x)$.



Copy the diagram neatly and sketch the curve $y = f'(x)$ on the same axes.

2

- d)
- If α and β are the roots of $2x^2 - 5x - 4 = 0$, form a quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.
- 4

Question 5

- a)
- Consider the function $f(x) = \frac{x}{4 - x^2}$
- (i)

State the domain of the function

1
- (ii)

Show that the function is an odd function

2
- (iii)

Show that the function is increasing throughout its domain.

3
- (iv)

Evaluate $\lim_{x \rightarrow \infty} f(x)$

1
- (v)

Sketch the graph of $y = f(x)$ showing all important features

2

b)

Prove by Mathematical Induction that $1 + 4 + 16 + 64 + \dots + 4^{n-1} = \frac{1}{3}(4^n - 1)$ for all positive integers n .

3

Question 6

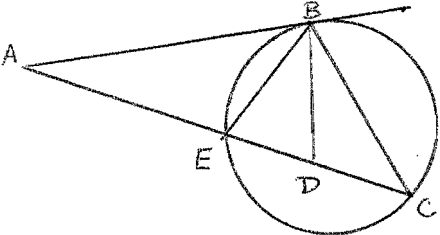
a) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0^\circ < x < 90^\circ$. 2

(ii) Hence or otherwise solve the equation $\sin x + \sqrt{3} \cos x = \sqrt{2}$ for $0^\circ \leq x \leq 360^\circ$ 2

b) Find the values of k for which the equation $2x^2 - 4x + k = 0$ has real roots 3

c) Simplify $\frac{3^{n+1} + 3^{n-1}}{3^{n-2} + 3^n}$ 2

d) AB is a tangent and AC is a secant of the circle. 3



BD bisects $\angle CBE$.
Let $\angle CBD = x$.
Prove that $\triangle ABD$ is isosceles.

Question 7

a) For which values of x is the curve $y = x^3 - 12x + 16$ decreasing **and** concave down? 3

b) Simplify $\frac{1}{2}(\cot \frac{\theta}{2} - \tan \frac{\theta}{2})$, expressing any trigonometric ratios in terms of θ 2

b) Anna wins \$20 000 in a lottery and decides to invest it in an account paying 10% p.a. compound interest, being compounded annually. She decides to withdraw \$2500 each year immediately after the annual interest has been credited to her account.

(i) After Anna makes her first withdrawal, how much remains in the account? 1

(ii) Let A_n = value of the investment after the n^{th} withdrawal. Show that $A_n = 25000 - 5000(1.1)^n$ 4

(iii) How many withdrawals of \$2500 will Anna be able to make? 2

Q1

a) $(2^4)^{4-x} = (2^{-3})^x$ ①

$$2^{16-4x} = 2^{-3x}$$

$$16-4x = -3x$$
 ①

$$16 = x$$
 ①

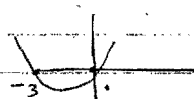
b) $\frac{x-6}{x} \geq 3$ $x \neq 0$

$$x^2 - 6x \geq 3x^2$$
 ①

$$0 \geq 2x^2 + 6x$$

$$2x^2 + 6x \leq 0$$

$$2x(x+3) \leq 0$$
 ①



$$-3 \leq x \leq 0$$
 ①

c) $y = (2x-4)^{-1/2}$

$$\frac{dy}{dx} = -\frac{1}{2}(2x-4)^{-3/2} \cdot 2$$

$$= -(2x-4)^{-3/2}$$

$$= \frac{-1}{(\sqrt{2x-4})^3}$$

$$= \frac{-1}{(\sqrt{6-4})^3}$$
 When $x=3$

$$= \frac{-1}{2\sqrt{2}}$$
 ①

d) A(4,6) B(13,5)

$$4 : -1$$

$$\left(\frac{4(13) - 1(4)}{4-1}, \frac{4(5) - 1(6)}{4-1} \right)$$

$$= \left(\frac{48}{3}, \frac{14}{3} \right)$$

$$= \left(16, \frac{14}{3} \right)$$

e) $-3 + 4 - 5 + 6$

$$= 2$$
 ①

1. d) Internal division
 $\left(\frac{56}{3}, \frac{26}{3} \right)$

= 1 mark only

12 marks each question

TOTAL/84

Q2

a) $y = 2x-1$, $m_1 = 2$

$$3x-2y+1=0$$
, $m_2 = \frac{3}{2}$ ①

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - \frac{3}{2}}{1 + 2(\frac{3}{2})} \right|$$
 ①

$$= \left| \frac{\frac{1}{2}}{4} \right|$$

$$= \frac{1}{8}$$

$$\theta = \tan^{-1}(\frac{1}{8}) \approx 7^\circ$$
 ①

b) $2 \sin \theta \cos \theta + \sqrt{3} \cos \theta = 0$

$$\cos \theta (2 \sin \theta + \sqrt{3}) = 0$$

$$\cos \theta = 0 \text{ or } \sin \theta = -\frac{\sqrt{3}}{2}$$

$$\theta = 90^\circ, 270^\circ, 240^\circ, 300^\circ$$

c) Let $u = x^2$

$$u^2 + 2u - 8 = 0$$
 ①

$$(u+4)(u-2) = 0$$
 ①

$$u = x^2 = -4$$

No soln

$$u = x^2 = 2$$

$$x = \pm \sqrt{2}$$
 ①

d) $f(x+h) = (x+h)^2 - 2(x+h)$

$$= x^2 + 2xh + h^2 - 2x - 2h$$
 ①

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$$
 ①

$$= \lim_{h \rightarrow 0} \frac{h(2x+h-2)}{h}$$
 ①

$$= 2x - 2$$

Q3

o) Ld of line from centre (0,0)

$$= \left| \frac{3(0) + 4(0) - k}{\sqrt{3^2 + 4^2}} \right|$$

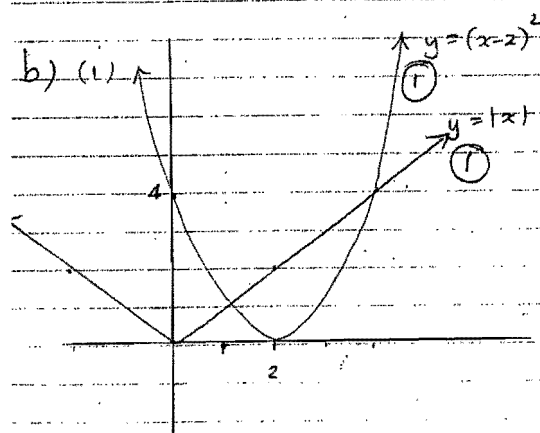
$$= \left| \frac{-k}{5} \right|$$

If line is a tangent,
then this distance = radius.

$$\left| \frac{-k}{5} \right| = 4$$

$$\frac{k}{5} = 4 \quad \text{or} \quad -\frac{k}{5} = 4$$

$$k = 20 \quad \text{or} \quad -20$$



(ii). Graphs intersect when

$$x = (x-2)^2$$

$$x = x^2 - 4x + 4$$

$$0 = x^2 - 5x + 4$$

$$0 = (x-4)(x-1)$$

$$x = 4, x = 1$$

Solu. to inequality is

$$x < 1, x > 4$$

$$c) y = x^3 + ax^2 + bx + c$$

$$y' = 3x^2 + 2ax + b$$

$$y'' = 6x + 2a$$

① both

Passes through (1,2)

$$2 = 1 + a + b + c$$

$$a + b + c = 1$$

Inflexion at (1,2)

Set $y'' = 0$

$$6(1) + 2a = 0$$

$$2a = -6$$

$$a = -3$$

①

Stat pt when $x = 4$ Set $y' = 0$

$$3(4)^2 + 2a(4) + b = 0$$

$$48 - 24 + b = 0$$

$$b = -24$$

①

$$y = x^3 - 3x^2 - 24x + c$$

Passes through (1,2)

$$2 = 1 - 3 - 24 + c$$

$$c = 28$$

①

Q4

$$a) \tan 56^\circ = \frac{50}{XB}$$

$$XB = \frac{50}{\tan 56^\circ} = 50 \cot 56^\circ$$

$$\tan 39^\circ = \frac{50}{YB}$$

$$YB = \frac{50}{\tan 39^\circ} = 50 \cot 39^\circ$$

$$XY^2 = YB^2 - BX^2 \text{ (Pythag)}$$

$$XY^2 = 50^2 \cot^2 39^\circ - 50^2 \cot^2 56^\circ$$

$$XY^2 = 50^2 (\cot^2 39^\circ - \cot^2 56^\circ)$$

①

$$b = XY = 50 \sqrt{\cot^2 39^\circ - \cot^2 56^\circ}$$

$$= 51.7 \text{ m}$$

①

$$b) \cos(A-B)$$

$$= \cos A \cos B + \sin A \sin B$$

①

$$\cos 15^\circ$$

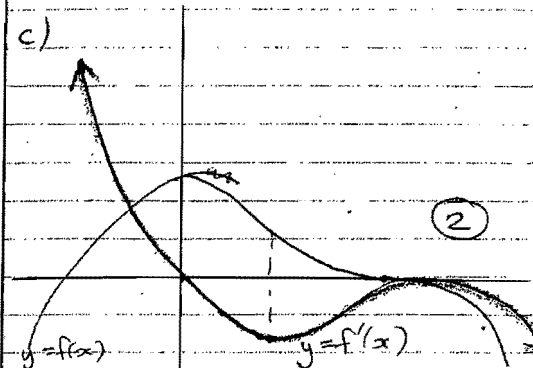
$$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

①

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

①



Last portion may lie
above or below $y = f(x)$

$$d) \alpha + \beta = \frac{5}{2}$$

$$\alpha \beta = -2$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha \beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha \beta}{\alpha \beta}$$

①

$$= \frac{\frac{25}{4} + 4}{-2}$$

$$= -\frac{41}{8}$$

①

$$\frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

①

Equation is

$$x^2 + \frac{41}{8}x + 1 = 0$$

①

Q5.

a) (i) $x \neq 2, -2$ ①

(ii) $f(-x) = \frac{-x}{4 - (-x)^2}$ ①

$$= \frac{-x}{4 - x^2}$$

$$= -\left(\frac{x}{4 - x^2}\right)$$

$$= -f(x) \quad \text{①} \quad \text{Fn is odd}$$

(iii) $f'(x) = \frac{(4 - x^2) \cdot 1 - x \cdot (-2x)}{(4 - x^2)^2}$ ①

$$= \frac{4 - x^2 + 2x^2}{(4 - x^2)^2}$$

$$= \frac{x^2 + 4}{(4 - x^2)^2} \quad \text{①}$$

Numerator & denominator are both always positive
(for ~~all~~ all x in domain) ①

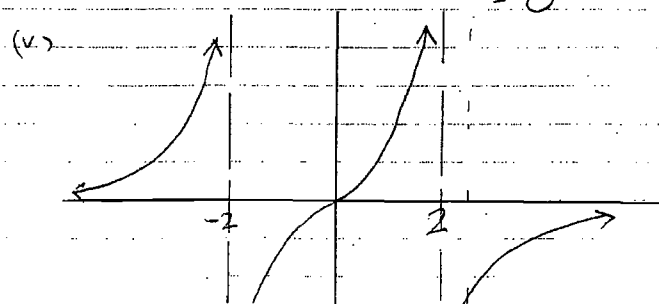
$\therefore f'(x) > 0$ for all x in domain

i.e. $f(x)$ is increasing

(iv) $\lim_{x \rightarrow \infty} \frac{x^2}{\frac{4}{x^2} - 1} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{\frac{4}{x^2} - 1}$

$$= \frac{0}{0 - 1}$$

$$= 0 \quad \text{①}$$



① for ↗

① for ↘

b) To prove $1 + 4 + 16 + 64 + \dots + 4^{n-1} = \frac{1}{3}(4^n - 1)$

for all pos. integers n

If $n = 1$ LHS = $4^{1-1} = 4^0 = 1$

RHS = $\frac{1}{3}(4^1 - 1) = \frac{1}{3}(3) = 1$

$\therefore$ Statement is true for $n = 1$ ①

Assume true for $n = k$

i. Assume $1 + 4 + 16 + 64 + \dots + 4^{k-1} = \frac{1}{3}(4^k - 1)$

Need to prove true for $n = k + 1$

ii. Need to prove:

$$1 + 4 + 16 + 64 + \dots + 4^{k-1} + 4^k = \frac{1}{3}(4^{k+1} - 1) \quad \text{①}$$

LHS = $\frac{1}{3}(4^k - 1) + 4^k$

by assumption

$$= \frac{1}{3}(4^k) - \frac{1}{3} + 4^k$$

$$= \frac{4}{3}(4^k) - \frac{1}{3}$$

$$= \frac{1}{3}(4(4^k) - 1)$$

$$= \frac{1}{3}(4^{k+1} - 1) \quad \text{①}$$

= RHS

$\therefore$ If statement is true for $n = k$, then it is also true for $n = k + 1$

Now statement is true for $n = 1$

$\therefore$ Also true for $n = 1 + 1 = 2$

" " " $n = 2 + 1 = 3$ etc

By induction, statement is true for all pos. integers n .

[-1 If conclusion is incomplete or incorrect]

Q6

a) (i) $R \sin(x + \alpha)$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

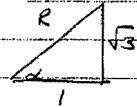
$$= \sin x + \sqrt{3} \cos x$$

$R \cos \alpha = 1$

$R \sin \alpha = \sqrt{3}$

$\cos \alpha = \frac{1}{2}$

$\sin \alpha = \frac{\sqrt{3}}{2}$



$R = 2$

$$\frac{R}{2} = 1, \quad \alpha = \tan^{-1} \sqrt{3} = 60^\circ$$

$$\sin x + \sqrt{3} \cos x = 2 \sin(x + 60^\circ)$$

(ii) $2 \sin(x + 60^\circ) = \sqrt{2}$

$\sin(x + 60^\circ) = \frac{\sqrt{2}}{2} \text{ (or } \frac{1}{\sqrt{2}})$

$$\begin{cases} 0^\circ \leq x \leq 360^\circ \\ 60^\circ \leq x + 60^\circ \leq 420^\circ \end{cases}$$

$x + 60^\circ = 45^\circ, 135^\circ, 405^\circ$

$x = 75^\circ, 345^\circ$

(Deduct 1 if -15° is left as a soln)

b) Real roots: $(\Delta \geq 0)$

$\Delta = (-4)^2 - 4 \cdot 2 \cdot k$

$= 16 - 8k \geq 0$

$-8k \geq -16$

$k \leq 2$

c) $\frac{3^{\frac{1}{2}}(3^{\frac{1}{2}} + 3^{-\frac{1}{2}})}{3^{\frac{1}{2}}(3^{-\frac{1}{2}} + 1)} = \frac{3^{\frac{1}{2}}}{1^{\frac{1}{2}}}$

$= 3$

d) If $\angle CBD = x$, then $\angle DBE = x$ (BD bisects $\angle CBE$)

$\angle ABE = \angle BCD$ ($\angle$ between tangent + chord = $\angle$ in alt. segment) ①

$\angle BDE = x + \angle DCB$ (ext $\angle$ of $\triangle DBC$ = sum of int. opp $\angle$ s) ①

$\angle ABD = x + \angle ABE$ (addition of adj. $\angle$ s)

but $\angle ABE = \angle DCB$ (proven above)

$\therefore \angle EPB = \angle ABD$ (addition of equal $\angle$ s)

$\therefore \triangle ADB$ is isosceles (2 equal $\angle$ s) ①

Q7

a) $y = x^3 - 12x + 16$
 $y' = 3x^2 - 12$
 $y'' = 6x$ ①

$y' < 0$ for $3(x-2)(x+2) < 0$
 $-2 < x < 2$ ①

$y'' < 0$ for $x < 0$

$y' < 0$ and $y'' < 0$ for $-2 < x < 0$ ①
↑ decreasing ↑ conc. down

b) c) (i) $\$20\,000(1.1) - 2500$ ①

b) $\frac{1}{2}(\frac{1}{t} - t) = \frac{1}{2} \cdot \frac{1-t^2}{t} = \frac{1-t^2}{2t} = \cot \theta$
[$t = \tan \frac{\theta}{2}$]

$$(ii) \quad A_1 = 20\,000(1.1) = 25000$$

$$A_2 = A_1 \times 1.1 = 25000$$

$$= 20\,000(1.1)^2 = 2500(1.1) = 25000$$

$$= 20\,000(1.1)^2 = 2500(1.1 + 1) \quad \text{develop sequence} \quad (1)$$

$$A_3 = A_2 \times 1.1 = 25000$$

$$= 20\,000(1.1)^3 = 2500(1.1^2 + 1.1) = 25000$$

$$= 20\,000(1.1)^3 = 2500(1.1^2 + 1.1 + 1)$$

$$A_n = 20\,000(1.1)^n = 2500(1.1^{n-1} + 1.1^{n-2} + \dots + 1.1 + 1) \quad (1)$$

geom. seq. $a=1$, $r=1.1$, n terms
 $S_n = \frac{a(r^n - 1)}{r - 1}$

$$= \frac{1 - (1.1)^n}{1.1 - 1}$$

$$= \frac{(1.1)^n - 1}{0.1} \quad (1)$$

$$A_n = 20\,000(1.1)^n = 2500 \times 10(1.1^n - 1)$$

$$= 20\,000(1.1)^n = 25000(1.1^n - 1) \quad (1)$$

$$= 20\,000(1.1)^n - 25\,000(1.1)^n + 25\,000$$

$$= 25\,000 - 5000(1.1)^n$$

iii) She can make withdrawals until $A_n = 0$

$$\text{If } A_n = 0, \text{ then } 25\,000 = 5000(1.1)^n$$

$$(1.1)^n = 5 \quad (1)$$

By trial & error (or logarithms), 16 withdrawals can be made. $\leftarrow (1)$

$$1.1^{16} = 4.6 \quad 1.1^{17} = 5.1 \quad \text{hence } 17$$